

Problem 1

i) $L^{clip}(10; \theta_k) := \tilde{E}_{s \sim d_{\pi_{\theta_k}}^{adv}} [L^{clip}(s, a)]$, $L^{clip}(s, a) := \min \{P_+(10) \cdot A^{adv}_{(s, a)}, clip(P_+(10), 1+\epsilon, 1-\epsilon) A^{adv}_{(s, a)}\}$
 $P_+(10) = \frac{\pi_{\theta}(1|s)}{\pi_{\theta_k}(1|s)}$ (新舊 policy 的機率比值)

ii) $\tilde{L}_{clip}(10; \theta_k) := clip(\frac{\pi_{\theta}(1|s)}{\pi_{\theta_k}(1|s)}, 1-\epsilon, 1+\epsilon) \cdot A^{adv}_{(s, a)}$

(sol) 兩個 objective function 最大的差異在於是否有取 min. 而在什麼情況下會導致這二個 objective function 產生不同的結果. 根據 Figure 1 分為 $A > 0$ 和 $A < 0$ 二種情況討論

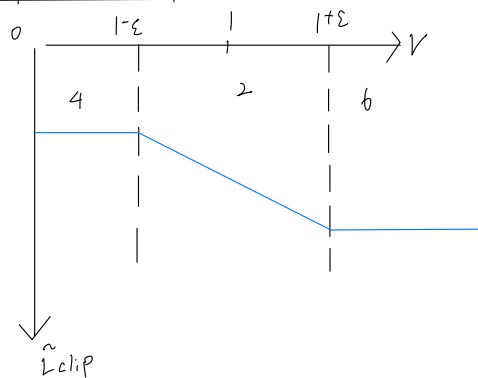
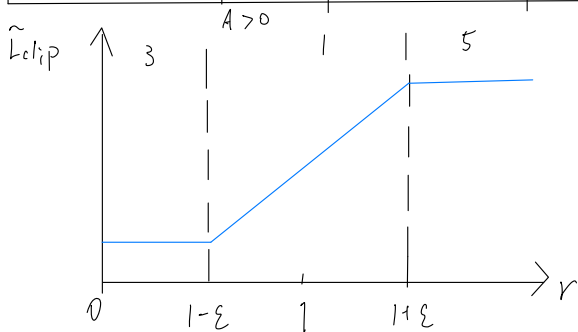
(I) 當 Advantage > 0 時:

$\Rightarrow A^{adv}_{(s, a)} > 0$, 代表在 state s 下, action a 的表現比平均好, 因此會想增加該動作的機率.
 在 $P_+(10) \in [1-\epsilon, 1+\epsilon]$ 和 $P_+(10) > 1+\epsilon$ 時, L_{clip} 和 $\tilde{L}_{clip}$ 所產生的結果是相同的, 但在 $P_+(10) < 1-\epsilon$ 時, 二者的結果開始產生差異.
 對於 L^{clip} 在 $P_+ < 1-\epsilon$, $\therefore A^{adv}_{(s, a)} > 0$ and $P_+ < 1-\epsilon$, 要取 min, 所以 Return Value 仍是 $P_+(10) \cdot A$.
 $\tilde{L}^{clip}$ 在 $P_+ < 1-\epsilon$ 單純使用 clip 強制歸 0 gradient, 所以 return value 是 $(1-\epsilon) \cdot A$.

(II) 當 Advantage < 0 時:

$\Rightarrow A^{adv}_{(s, a)} < 0$, 代表在 state s 下, action a 的表現比平均差, 因此會想減少該動作的機率.
 在 $P_+(10) \in [1-\epsilon, 1+\epsilon]$ 和 $P_+(10) < 1-\epsilon$, L_{clip} 和 $\tilde{L}_{clip}$ 的結果相同, 但當 $P_+(10) > 1+\epsilon$, 結果不同.
 對於 L^{clip} 在 $P_+ > 1+\epsilon$, $\therefore A^{adv}_{(s, a)} < 0$ and $P_+ > 1+\epsilon$, 取 min 所以 Return value 仍是 $P_+(10) \cdot A$.
 $\tilde{L}^{clip}$ 在 $P_+ > 1+\epsilon$ 單純 clip 會歸 0 gradient, 讓 P_+ 停在 $1+\epsilon$, 因此 Return value 為 $(1+\epsilon) \cdot A$.
 How $\tilde{L}^{clip}$:

	$P_+(10) > 0$	A_c	Return Value of min	Objective is clipped	Sign of objective	Gradient
1	$P_+(10) \in [1-\epsilon, 1+\epsilon]$	+	$P_+(10) \cdot A_c$	no	+	✓
2	$P_+(10) \in [1-\epsilon, 1+\epsilon]$	-	$P_+(10) \cdot A_c$	no	-	✓
3	$P_+(10) < 1-\epsilon$	+	$(1-\epsilon) A_c$	yes	+	0
4	$P_+(10) < 1-\epsilon$	-	$(1-\epsilon) A_c$	yes	-	0
5	$P_+(10) > 1+\epsilon$	+	$(1+\epsilon) A_c$	yes	+	0
6	$P_+(10) > 1+\epsilon$	-	$(1+\epsilon) A_c$	yes	-	0



Problem 2.

(a.)

For the case $S \neq S_t$, $V_t(S)$ update should be $V_{t+1}(S) = V_t(S)_*$

將 $V_{t+1}(S_t) = V_t(S_t) + \alpha_t(S_t) (Y_{t+1} + \gamma V_t(S_{t+1}) - V_t(S_t))$ 整理成 Stochastic Approximation 公式.

$\Rightarrow V_{t+1}(S_t) = (1 - \alpha_t(S_t)) V_t(S_t) + \alpha_t(S_t) (Y_{t+1} + \gamma V_t(S_{t+1})) \stackrel{a.s.}{\rightarrow} V_t(S_t) = 0$ 等. $V_{t+1}(S) = V_t(S)_*$

(b.)

(i). Recall TD(0) algorithm equation: $V_{t+1}(S_t) = V_t(S_t) + \alpha_t(S_t) \cdot (Y_{t+1} + \gamma V_t(S_{t+1}) - V_t(S_t))$
 $\Rightarrow$ 逆用同樣方法: 將 $V_t^*(S_t) : V_{t+1}(S_t) - V_t^*(S_t) = V_t(S_t) - V_t^*(S_t) + \alpha_t(S_t) \cdot (Y_{t+1} + \gamma V_t(S_{t+1}) - V_t(S_t))$, 整理成 Stochastic approximation 公式形式:

$\Rightarrow V_{t+1}(S_t) - V_t^*(S_t) = (1 - \alpha_t(S_t)) \cdot (V_t(S_t) - V_t^*(S_t)) + \alpha_t(S_t) \cdot (Y_{t+1} + \gamma V_t(S_{t+1}) - V_t(S_t))$, 對 $t \rightarrow \infty$ 可得

$\alpha_t(S_t) = \alpha_t(S_t)$, $W_t(S_t) = V_t(S_t) - V_t^*(S_t)$, $\xi_t(S_t) = Y_{t+1} + \gamma V_t(S_{t+1}) - V_t(S_t)$

(ii). For SASE, the value is not updated

$W_{t+1}(S) = (1 - \alpha_t(S)) W_t(S) + \alpha_t(S) \cdot \xi_t(S)$, $W_{t+1}(S) = W_t(S)$

, so $W_t(S) = V_t(S) - V_t^*(S)$, $\xi_t(S) = 0$.

(c.) $\mathcal{P}_{\text{over}}(C_1)$ and $\mathcal{C}(C_2)$

(i). For $\mathcal{C}(C_1)$ $|E[\xi_t(S)|H_t]| \leq \rho \|W_t\|_\infty$

$\Rightarrow |E[\xi_t(S)|H_t]| = |E[Y_{t+1} + \gamma V_t(S_{t+1}) - V_t(S_t) | H_t]|$
 $= |E[Y_{t+1} + \gamma V_t(S_{t+1}) - \gamma V_t^*(S_{t+1}) | H_t]|$
 $\leq \gamma \max_t |V_t(S_t) - V_t^*(S_t)|$
 $= \gamma \|V_t(S_t) - V_t^*(S_t)\|_\infty$
 $= \gamma \|W_t\|_\infty$, $\gamma = \rho$ as discount factor.

(ii). For $\mathcal{C}(C_2)$ $|V_t(S_t)|H_t| \leq C(1 + \|W_t\|_\infty)$

1. $\xi_t(S_t) = Y_{t+1} + \gamma V_t(S_{t+1}) - V_t(S_t)$, 要證其 W_t 因此同時減 $\gamma V_t^*(S_{t+1})$

$\Rightarrow \xi_t(S_t) = Y_{t+1} + \gamma V_t^*(S_{t+1}) - V_t^*(S_t) + \gamma (V_t(S_{t+1}) - V_t^*(S_{t+1}))$
 $= (Y_{t+1} + \gamma V_t^*(S_{t+1}) - V_t^*(S_t)) + \gamma W_t(S_{t+1})$

2. $\because$ By definition of variance, $V(X) = E[X^2] - (E[X])^2$

$\Rightarrow V[\xi_t(H_t)] = E[\xi_t^2(H_t)] + (E[\xi_t(H_t)])^2$, $E[\xi_t(H_t)]$ is a constant

So let's focus on $E[\xi_t^2(H_t)]$.

$E[\xi_t^2(H_t)] = E[(Y_{t+1} + \gamma V_t^*(S_{t+1}) - V_t^*(S_t) + \gamma W_t(S_{t+1}))^2]$

3. Recall the Cauchy-Schwarz inequality: $(a+b)^2 \leq 2a^2 + 2b^2$

Let $a = Y_{t+1} + \gamma V_t^*(S_{t+1}) - V_t^*(S_t)$, $b = \gamma W_t(S_{t+1})$

$\Rightarrow E[\xi_t^2(H_t)] \leq 2E[(Y_{t+1} + \gamma V_t^*(S_{t+1}) - V_t^*(S_t))^2] + 2\gamma^2 E[W_t^2(S_{t+1})]$, $\because Y_{t+1} + \gamma V_t^*$ 是已知的, 因此設 $C_{\text{noise}} = 2E[(Y_{t+1} + \gamma V_t^*(S_{t+1}) - V_t^*(S_t))^2]$

而根據 L_∞ 的定義: $\|W_t\|_\infty = \max_{i \in S} |W_t(i)|$, $\therefore W_t(S_{t+1}) \leq \|W_t\|_\infty$

$\therefore E[\xi_t^2(H_t)] \leq C_{\text{noise}} + 2\gamma^2 \|W_t\|_\infty^2$

4. For $V[\xi_t(H_t)] \leq C + 2\gamma^2 \|W_t\|_\infty^2$, 舉例來說 $A + B\|W_t\|_\infty^2 \leq C(1 + \|W_t\|_\infty) \leq C(1 + \|W_t\|_\infty)$, 只要 C 足夠大便會成立.

$\Rightarrow V[\xi_t(H_t)] \leq C(1 + \|W_t\|_\infty)$ 成立

(d) 根據 stochastic approximation Theorem, 需要滿足 $\mathcal{C}(C_1)$, $\mathcal{C}(C_2)$, $\mathcal{C}(C_3)$ 條件, $W_t \rightarrow 0$ as $t \rightarrow \infty$ with probability 1.

在 (b) 時, 我們便已證得 $W_t = V_t(S_t) - V_t^*(S_t)$, $W_t = 0$, $t \rightarrow \infty$, so $V_t(S_t) = V_t^*(S_t)$ when $t \rightarrow \infty$

150 L.)

② 限制条件: $\frac{1}{2}(1-\theta_k)H_k(1-\theta_k) - \frac{1}{2} \leq 0$: θ 与 θ_k 同时存在不能超过 1

定义 Lagrangian (约束目标函数和约束中保持): $L(\theta, \lambda) = -(\nabla_{\theta} L_{\theta}(\theta))^\top (\theta - \theta_*) + \lambda (\frac{1}{2} \|\theta - \theta_*\|^2 + H_{\theta_*}(\theta - \theta_*)) - \zeta$

$\Rightarrow \lambda$: 当违反 λ 限制条件, 微罚的倍率.
 $D(\lambda)$ (duality function) = $\min_{0 \leq \lambda} L(0, \lambda)$, 代表不论 λ 取何值, 在给定一个固定的倍率 λ , 所能得到的最小微罚.

而 duality problem 所要寻找的是找到一个 λ 使得 λ 的总对偶收入 $\lambda = \sum \lambda_i x_i^0$ 与 λ^0 相等。
 例如, 能存在一个主问题约束限制条件成立 $\Rightarrow$ Solving duality (即原问题最优化的解 = duality problem 最优化的解)

$$L_{\theta, \lambda} = -(\nabla_{\theta} L_{\theta, \lambda}(\theta) |_{\theta = \theta_k})^T (\theta - \theta_k) + \lambda \left(\frac{1}{2} (\theta - \theta_k)^T H_{\theta_k} (\theta - \theta_k) - \zeta \right)$$

$$J(\lambda) = \min_{\theta \in \mathbb{R}} L(\theta, \lambda)$$

$\Rightarrow P(\lambda)$ 是要找出 θ 使 $L(\theta, \lambda)$ 值最小.

$$\frac{\partial}{\partial \theta} L(\theta, \lambda) = \frac{\partial}{\partial \theta} \left\{ -(\nabla_{\theta} L(\theta_k | \theta)) |_{\theta = \theta_k}^T \cdot (\theta - \theta_k) \right\} + \frac{\partial}{\partial \theta} \left\{ \lambda \left(\frac{1}{2} (\theta - \theta_k)^T \right) H_{\theta_k} (\theta - \theta_k) - \xi \right\}$$

①: $\lim_{t \rightarrow 0} \nabla_{\theta} L(\theta; t) / \theta \cdot \theta_k = f$. $\therefore f$ 与 θ 無關, 是一個定值

$$\therefore \frac{1}{j\omega} \left[- \int \cdot (10 - \theta_k) \right] = g \quad (\because V_{\text{ising}} \text{ 矢磁率の微分公式, } \frac{d}{dx} (a^T x) = a)$$

②: Recall $\frac{d}{dx} (x^T a x) = 2ax$

$$\frac{d}{d\theta} \left\{ \lambda \frac{1}{2} (\theta - \theta_k)^T H_{\theta_k} (\theta - \theta_k) - \xi \right\} = \frac{\lambda}{2} \cdot (2 H_{\theta_k} (\theta - \theta_k))$$

$$\Rightarrow \frac{d}{d\theta} L(\theta, \lambda) = -(\nabla_{\theta} L_{\theta_k}(\theta) | \theta, \theta_k) + \lambda H_{\theta_k}(\theta - \theta_k) \stackrel{L.f.}{=} 0.$$

$$\Rightarrow \theta = \frac{H_{\theta\theta}^{-1}(\nabla_{\theta} L_{\theta}(v) | v = v_{\theta})}{\lambda} + \theta_{\theta}. \quad \text{ist die Lagrangian-Multiplikation.}$$

$$\begin{aligned} \mathcal{L}(\gamma, \lambda) &= -(V_b L_{b\gamma} |b\rangle \langle b| v_k) \left(\frac{H_{b\gamma}^+}{\lambda} (V_b L_{b\gamma} |b\rangle \langle b| v_k) \right) + \lambda \left(\frac{H_{b\gamma}^+}{\lambda} (V_b L_{b\gamma} |b\rangle \langle b| v_k) \right)^T \cdot H_{b\gamma} \left(\frac{H_{b\gamma}^+}{\lambda} (V_b L_{b\gamma} |b\rangle \langle b| v_k) - \delta \right) \\ &= \left(-(V_b L_{b\gamma} |b\rangle \langle b| v_k)^T + \frac{1}{\lambda} (V_b L_{b\gamma} |b\rangle \langle b| v_k)^T \right) \cdot \left(\frac{H_{b\gamma}^+}{\lambda} (V_b L_{b\gamma} |b\rangle \langle b| v_k) \right) - \lambda \delta. \end{aligned}$$

$$= \frac{-1}{2\lambda} (\nabla_{\theta} L_{\theta_k}(\theta) / \nabla_{\theta} v_k)^T \cdot H_{\theta_k}^{-1} (\nabla_{\theta} L_{\theta_k}(\theta) / \nabla_{\theta} v_k) - \lambda \zeta_{\theta_k} = \mathcal{P}_{\lambda}(\theta)$$

$\lambda^{\pm} := \arg \max_{\lambda \geq 0} P_{\lambda}(\lambda)$, 要找到 λ 的极值对 $P_{\lambda}(\lambda)$ 而言, 对 $P_{\lambda}(\lambda)$ 求导数.

$$\frac{dP_{(A)}}{d\lambda} = \frac{1}{2\lambda^2} (\langle \nabla_\theta L_{\theta_k}(b) | \nabla_\theta \psi_k \rangle)^T \cdot H_{\theta_k}^{-1} (\langle \nabla_\theta L_{\theta_k}(b) | \nabla_\theta \psi_k \rangle) - \zeta \stackrel{\text{let}}{=} 0$$

$$\Rightarrow \lambda^* = \left(\left(\frac{1}{2\beta} (\nabla_{\theta} L_{\theta_k}(\theta)) /_{\theta=\theta_k} \right)^T \cdot H_{\theta_k}^{-1} (\nabla_{\theta} L_{\theta_k}(\theta)) /_{\theta=\theta_k} \right)^{\frac{1}{2}}$$

(b.) $\theta^* = \arg \min_{\theta} L(\theta, \lambda^*)$. Show that $\theta^* = \theta_k + \alpha H_{\theta k}^{-1} \nabla_{\theta} L(\theta_k, \lambda^*) / \|\nabla_{\theta} L(\theta_k, \lambda^*)\|$.

$$\theta^* = \frac{1}{\lambda} \cdot H_{\theta\theta}^{-1} \cdot (V_{\theta} L_{\theta\theta} | \theta; \psi_k) + \psi_k, \quad \lambda = \frac{1}{\lambda^*} = \left(\frac{1}{2} (V_{\theta} L_{\theta\theta} | \theta; \psi_k)^T \cdot H_{\theta\theta}^{-1} \cdot (V_{\theta} L_{\theta\theta} | \theta; \psi_k) \right)^{-\frac{1}{2}}$$